



Unit 4 · Outcome 3 · Scientific Investigation

Accuracy, Precision & Validity of Measurements

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

When you generate your own data in an environmental investigation, the markers — and the science — only trust your numbers if you can describe how good they are. This module unpacks the five words VCAA expects you to use precisely: accuracy, precision, repeatability, reproducibility and validity.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



MAP The five words, in one breath

Before the detail, here's the whole dot point at a glance. Two questions are about a set of measurements (accuracy, precision). Two are about whether other people/conditions get the same answer (repeatability, reproducibility). One is about whether you measured the right thing at all (validity).

How close a measurement (or their mean) is to the true value. Hurt by systematic error.

How close repeated measurements are to each other (their spread). Hurt by random error.

Same answer when you repeat it under the same conditions (same method, gear, place, short time).

Same answer when conditions change — a different person, instrument, lab or method.

Did you actually measure what you set out to measure? Controlled variables + accurate method.

Tip: scroll the outline ► each term gets its own worked Mullawa Creek example.

✓ TIP

The one-sentence test. Accuracy asks “is it right?” · Precision asks “is it consistent?” · Repeatability & reproducibility ask “does it stay the same when I (or someone else) do it again?” · Validity asks “am I even measuring the right thing?”

A Accuracy — closeness to the true value

How close a measurement is to the true (accepted) value of the quantity being measured.

Accuracy is judged against a reference. You literally cannot say a number is accurate unless you know — or can reasonably accept — what the right answer is. At Mullawa Creek that reference is the calibration standard at 8.20 mg/L. After taking five readings of that standard, a group works out the mean and compares it to 8.20.

Group	Five readings of the 8.20 standard (mg/L)	Mean	Accurate?
A · fresh probe	8.18, 8.22, 8.19, 8.21, 8.20	8.20	Yes — mean $\approx$ true value
B · skipped calibration	8.71, 8.69, 8.72, 8.70, 8.68	8.70	No — mean is +0.50 too high

Group B's probe wasn't zeroed, so every reading is shifted up by about 0.50 mg/L. That consistent one-direction shift is a systematic error — the classic enemy of accuracy. Notice their readings are beautifully consistent with each other... and consistently wrong.

⚠ WATCH OUT

Averaging won't save Group B. Taking more readings tightens the average but it stays parked at 8.70. A systematic error is not reduced by repeating — only by fixing the cause (re-calibrate, re-zero, correct the offset).

B Precision — closeness of repeats to each other

How close a set of measurement values are to one another. Precision gives no indication of how close those values are to the true value.

Precision is about spread, not correctness. A tight cluster of readings is precise; a scattered set is imprecise. The usual culprit behind poor precision is random error — small, unpredictable fluctuations that push readings up and down (probe drift, ripples on the water, you reading the scale a hair differently each time).

Group	Five readings of the 8.20 standard (mg/L)	Spread	Precise?
A · fresh probe	8.18, 8.22, 8.19, 8.21, 8.20	± 0.02	Yes — tight cluster
C · ageing probe	7.9, 8.6, 8.0, 8.5, 8.1	± 0.35	No — scattered

Group C is interesting: their mean is 8.22 — almost spot-on the true value — yet no single reading is trustworthy because they're all over the place. They got an accurate average by luck of cancellation, but their measurements are imprecise. This is exactly why repeating measurements and averaging helps with random error: the highs and lows tend to cancel.

• **NOTE**

One reading can't be "precise". Precision is a property of a set. If a question gives you a single value, you can comment on resolution (how fine the scale is) but not precision until it's repeated.

SCENE 2

LAB The calibration bench — take readings, watch quality emerge

This is the whole dot point in one toy. Pick a scenario, then hit Take reading a few times. Each shot lands on the 3D target and drops a dot on the number line below. The gauges compute the mean ($\rightarrow$ accuracy vs the 8.20 true value) and the spread ($\rightarrow$ precision) live.

Tight + centred $\rightarrow$ accurate & precise. The goal.

Tight but shifted $\rightarrow$ precise, not accurate. Systematic error.

Scattered around centre $\rightarrow$ accurate-ish mean, not precise. Random error.

DIAGRAM

Accuracy vs precision — the distinction examiners love

If you take one idea from this module, take this: accuracy and precision are independent. A measurement can be any combination of the two. The target picture below is worth memorising — it shows up in some form in almost every set of practice exams.

	Accurate	Not accurate
Precise	Ideal. Tight cluster on the bullseye. Calibrated probe.	Dangerous. Tight cluster, wrong spot $\rightarrow$ systematic error. Looks trustworthy but isn't.
Not precise	Lucky. Scattered, but the mean lands near true $\rightarrow$ random error, big averages needed.	Useless. Scattered and off-centre $\rightarrow$ both errors present.

 WATCH OUT

Trap. "The results were consistent, so they're accurate." Wrong. Consistency is precision. Group B was perfectly consistent and 0.50 mg/L off. Consistency only becomes accuracy once you've checked against a true/accepted value.

✓ TIP

How to improve each. Improve accuracy → calibrate instruments, remove zero-offsets, correct for parallax, use a reference standard. Improve precision → repeat and average, control conditions, use higher-resolution instruments, reduce environmental fluctuation.

C The two errors behind it all

Accuracy and precision each have a matching error type. Sorting causes into the right bucket is a fast way to lock this in — and it's a common 2–3 mark exam task.

A consistent shift in one direction. Same size every time. Not reduced by repeating. Fix the cause.

Unpredictable scatter, both directions. Averages out over many repeats. Reduced by repeating & averaging.

D Repeatability — same you, same conditions

The closeness of agreement between successive measurements of the same quantity, carried out under the same conditions of measurement — same method, same instrument, same observer, same location, repeated over a short period.

Repeatability is precision under locked-down, identical conditions. At Mullawa Creek: Priya takes five DO readings at the same sampling point, with the same probe, within ten minutes. If those five readings agree closely, her measurement is repeatable. It's the first, cheapest reliability check — and it's entirely within one student's control.

- Same observer — Priya does all five.
- Same instrument — the one calibrated probe.
- Same method & location — identical depth, identical technique, same spot.
- Short timeframe — before the creek's conditions can drift.

SCENE 3

E Reproducibility — change something, still agree

The closeness of agreement between measurements of the same quantity, carried out under changed conditions of measurement — e.g. a different observer, different instrument, different method, different location, or different time.

Reproducibility is the tougher, more convincing test. At Mullawa Creek: a second group, using a different probe on a different day, returns to the same point and gets the same DO. If independent conditions still agree, confidence in the result jumps — this is the everyday logic of peer review and replication across science.

What changes?	Repeatability	Reproducibility
Observer	Same	Can differ
Instrument	Same	Can differ
Method	Same	Can differ
Location / time	Same place, short time	Can differ
Tells you...	Your own technique is consistent	The finding holds up independently

⚠ WATCH OUT

The #1 mix-up. Repeatability = same conditions. Reproducibility = changed conditions. Memory hook: re P eat = same P erson/place; re P ro D uce = D ifferent set-up.

F Sort the quality

◇ ACTIVE RECALL

Each statement below describes one of the five qualities. Drag it to its bin — this is the exact skill behind "identify/justify which quality is being described" questions.

G Validity — did you measure the right thing?

A measurement or investigation is valid if it actually measures what it claims to / sets out to measure, and the evidence genuinely supports the conclusion drawn.

Validity is the big-picture judgement that sits above the other four. You can have accurate, precise, repeatable, reproducible numbers — and still answer the wrong question. Two layers to think about:

The instrument and technique measure the intended quantity. A DO probe reads dissolved oxygen — not temperature, not "general water health". Using a turbidity tube to claim DO would be invalid, however precisely you read it.

The independent variable is the only thing changing the dependent variable. Controlled variables (depth, time of day, technique) are held constant so any change in DO can be attributed to distance from the drain — not a confounder.

At Mullawa Creek, validity is threatened if — say — Group 1 sampled at 8 a.m. and Group 2 at 3 p.m. Photosynthesis raises DO through the day, so "distance from drain" gets tangled with "time of day". The data might be accurate and precise, but the comparison is invalid because a confounding variable wasn't controlled.

⚠ WATCH OUT

Validity ≠ precision/reliability. You can reliably and precisely measure the wrong thing. A consistently mis-zeroed probe gives precise, repeatable, reproducible numbers that are still invalid for stating the true DO. Precision is about the numbers; validity is about whether they answer your question.

• NOTE

How accuracy feeds validity. Valid conclusions need accurate measurements and controlled variables. That's why these five terms aren't a list — they're a chain: control the method → measure accurately & precisely → confirm by repeating & reproducing → then your conclusion is valid.

📝 Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you how much to write. Tap a card to see what the marker is actually looking for on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

P·E·L Build a band-6 answer, one move at a time



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Strong science responses follow Point → Evidence → Link. Draft each part below; the keyword meter lights up as you include the language markers worth marks. Then reveal a model to compare.

Worked answer — weak vs strong

Same question, two responses. The annotations show exactly where marks are won and lost.

Practice questions — 22 marks

★ PRACTICE & SELF-MARK

Attempt each in the box, reveal the marking guide, then self-mark honestly. Your tally feeds the dial in the corner and saves automatically.

★ Key takeaways

♦ KEY TAKEAWAY

Lock these in before the exam Accuracy = closeness to the true value; judged against a reference; ruined by systematic error; not fixed by repeating. Precision = closeness of repeats to each other (spread); says nothing about correctness; ruined by random error; improved by repeating & averaging. Accurate and precise are independent — memorise the four-quadrant target. Repeatability = same conditions (same person/gear/place, short time). Reproducibility = changed conditions (different person/gear/method/lab). Validity = measuring the right thing with controlled variables, so the evidence genuinely supports the conclusion. You can precisely measure the wrong thing. Use the exact VCAA wording and always anchor accuracy to a true/accepted value — that's where the marks live.

◆ NOTE

Real-world anchor. In Australia, measurement traceability runs back to the National Measurement Institute (NMI), which maintains the national standards instruments are calibrated against. EPA Victoria's water-quality monitoring relies on exactly this chain — calibrated probes, repeated readings, and reproducible methods across sites — so that a DO figure from one creek can be validly compared with another.

✦ RECAP / CHEAT SHEET

Digital Vector · VCE Environmental Science Units 3 & 4 · U·O3 Scientific Investigation · KK04. Aligned to the VCAA Environmental Science Study Design 2022–2026. Mullawa Creek is a fictional teaching scenario. Australian English.

SCENE 1

Accuracy, Precision & Validity of Measurements

When you generate your own data in an environmental investigation, the markers — and the science — only trust your numbers if you can describe how good they are. This module unpacks the five words VCAA expects you to use precisely: accuracy, precision, repeatability, reproducibility and validity.

• NOTE

Where this sits. KK01–KK03 got your investigation designed and your primary data generated. KK04 is the quality-control checkpoint: how good are the measurements you just made? Get this language right and it pays off everywhere — your method, your discussion, your evaluation of limitations (KK09), and several guaranteed exam marks.

◆ NOTE

Running scenario — Mullawa Creek. All chapter examples follow a Year 12 student-designed investigation at Mullawa Creek: testing how distance downstream from a stormwater drain affects water quality. Students measure dissolved oxygen (DO) with an electronic probe and turbidity with a turbidity tube. Before sampling, every probe is checked against a calibration standard with a known DO of 8.20 mg/L — our true value.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.